VMware ESXI Virtualization Lab

Virtual Machine Deployment Guide

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Project Overview

This document provides a step-by-step guide for creating and configuring a virtual machine using VMware ESXI. It includes screenshots, explanations, and configuration details for each stage of the deployment process.

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Introduction

Purpose

This document provides a detailed walkthrough of creating and configuring a virtual machine using **VMware ESXI 7.0 U2**. The guide documents each stage of the deployment process with screenshots and explanations to demonstrate practical knowledge of virtualization technologies.

Project Objectives

- Understand the VMware ESXI interface.
- Create a new virtual machine from scratch.
- Configure virtual hardware components.
- Install a guest operating system.
- Verify successful virtual machine deployment.
- Develop hands-on skills in virtualization and infrastructure management.

Scope

This guide focuses on deploying a Windows virtual machine in VMware ESXI. It covers VM creation, operating system selection, storage configuration, hardware customization, ISO mounting, operating system installation, and verification of successful deployment.

Prerequisites

Before creating the virtual machine, ensure the following requirements are met:

Hardware Requirements

- A system with VMware ESXI 7.0 U2 installed.
- Minimum 8 GB RAM (recommended).
- Sufficient storage space for the virtual machine.
- Stable network connectivity.

Software Requirements

- VMware ESXi Web Client access.
- Windows 10 (64-bit) ISO image.
- Web browser (Google Chrome, Microsoft Edge, or Mozilla Firefox).

Knowledge Requirements

- Basic understanding of virtualization concepts.
- Familiarity with operating systems.
- Basic networking knowledge.

Lab Environment

Host Environment

The virtual machine was created and configured using a VMware ESXi server provided for hands-on DevOps training.

Environment Details

Component	Details
Hypervisor	VMware ESXi 7.0 U2
Guest Operating System	Microsoft Windows 10 (64-bit)
Deployment Method	Create a New Virtual Machine
Management Interface	VMware ESXi Web Client
Installation Media	Windows 10 ISO Image

Purpose of the Lab

This lab was designed to provide practical experience in:

- Deploying virtual machines.
- Configuring virtual hardware.
- Managing virtual storage.
- Configuring virtual networking.
- Installing guest operating systems.
- Understanding virtualization workflows used in DevOps environments.

Step 1: Select Creation Type

Objective

Begin the virtual machine creation process by selecting the appropriate deployment method.

Procedure

1. Click **Create/Register VM**.
2. Select **Create a new virtual machine**.
3. Click **Next** to continue.

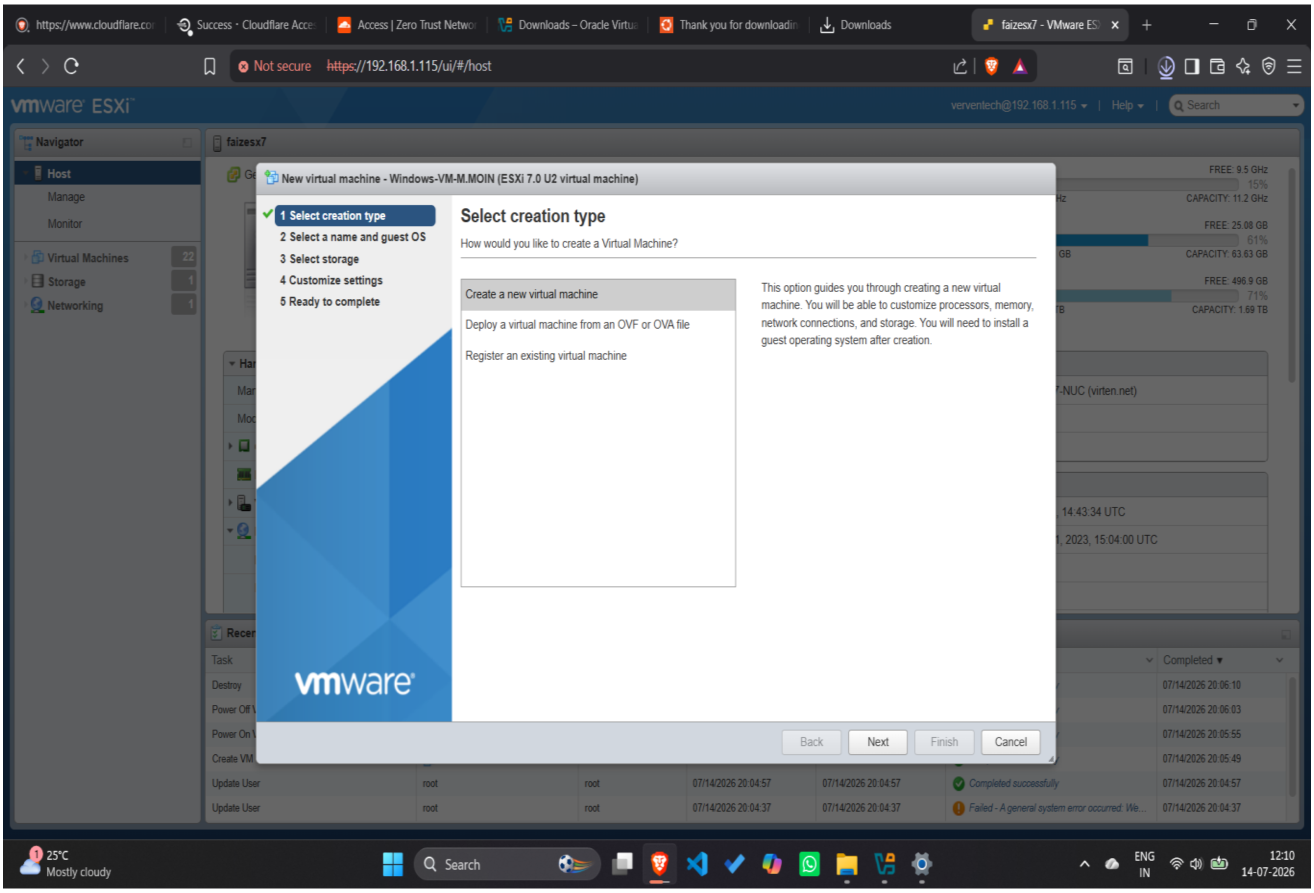


Figure 1: Selecting the virtual machine creation type.

Step 2: Configure VM Name and Guest Operating System

Objective

Assign a unique virtual machine name and select the guest operating system.

Procedure

1. Enter a unique virtual machine name.
2. Select **Windows** as the Guest OS Family.
3. Select **Microsoft Windows 10 (64-bit)** as the Guest OS Version.
4. Click **Next**.

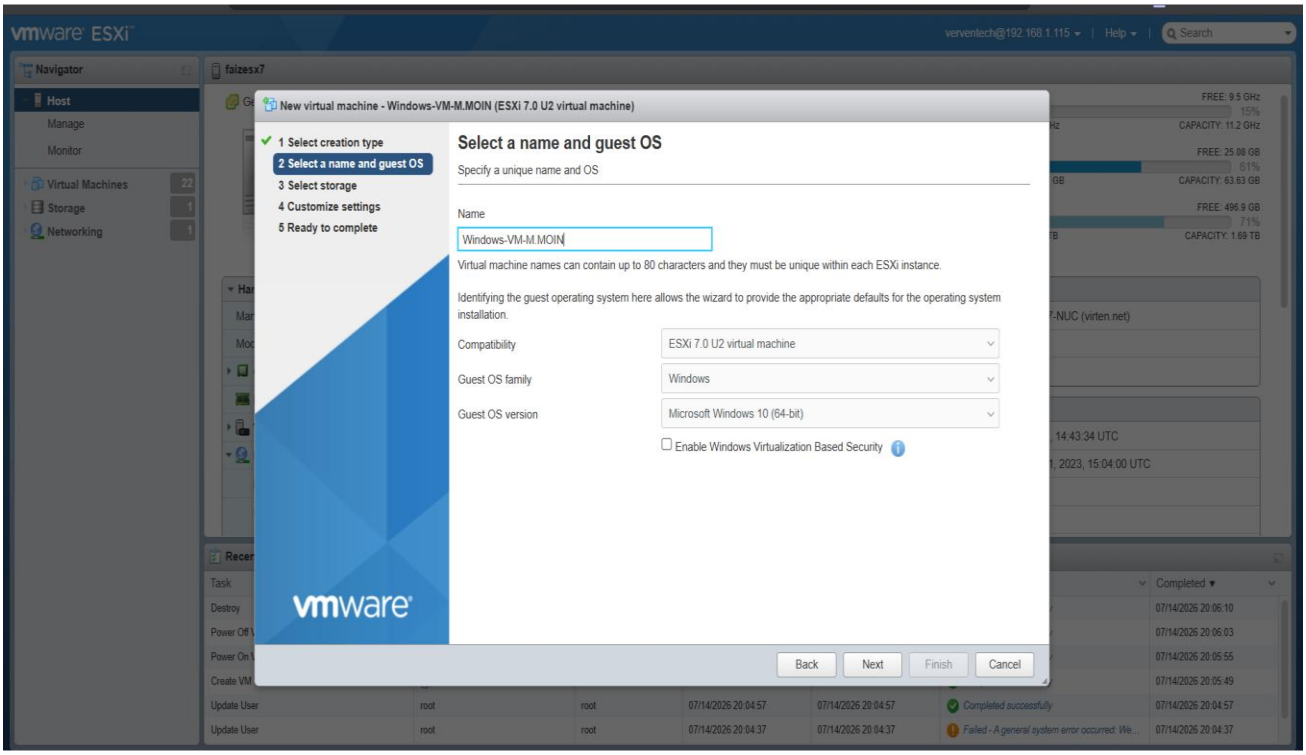


Figure 2: Configuring the VM name and guest operating system.

Step 3: Select Storage

Objective

Select the datastore where the virtual machine configuration files and virtual disks will be stored.

Procedure

1. Navigate to the **Select Storage** page of the VMware ESXi wizard.
2. Choose the **Standard** storage option.
3. Select the available datastore (**datastore1**) from the list.
4. Verify that sufficient storage capacity is available for the virtual machine.
5. Click **Next** to proceed.

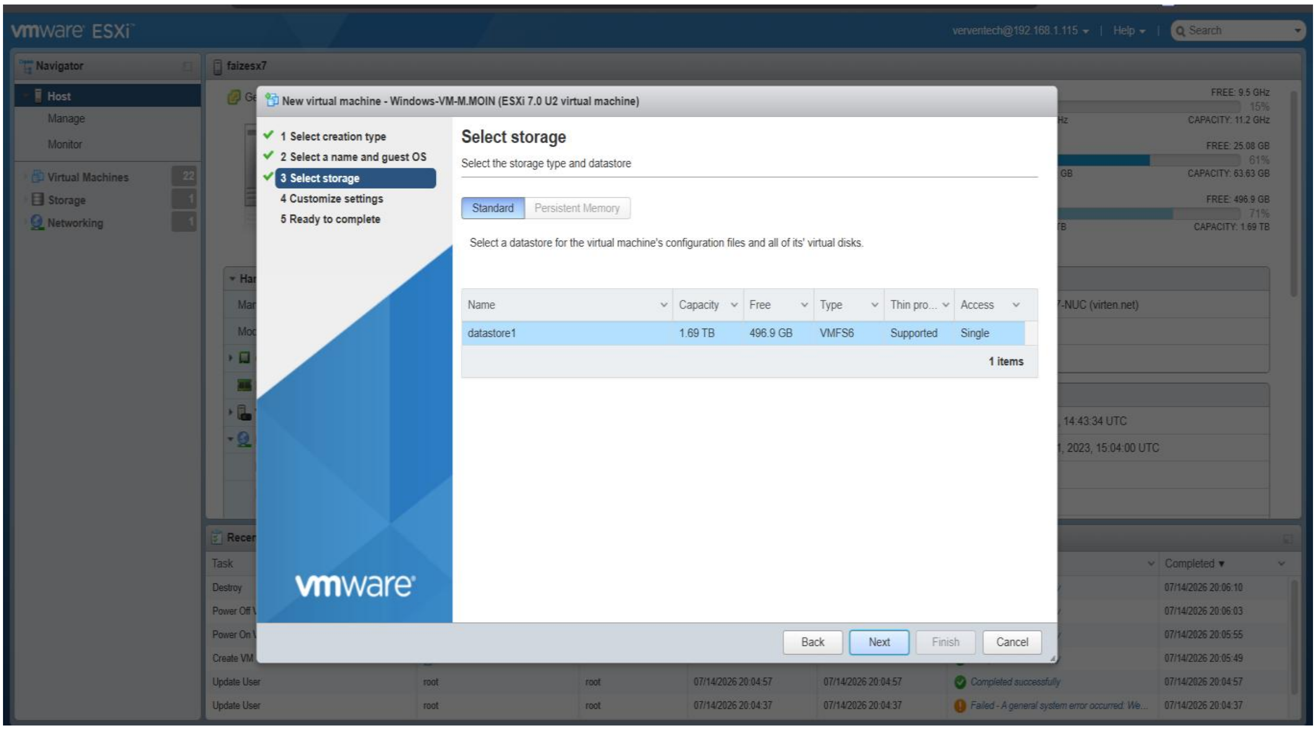


Figure 3: Selecting the datastore for virtual machine storage.

Note: A datastore is the storage location where VMware ESXi stores virtual machine configuration files, virtual disks, snapshots, and other VM-related data. Selecting a datastore with sufficient free space helps ensure reliable virtual machine deployment and future expansion.

Step 4: Customize Settings

Objective

Configure the virtual machine hardware settings and attach the installation media before deploying the virtual machine.

Procedure

1. Open the **Customize Settings** page.
2. Configure the virtual machine with:
    - **CPU:** 2 vCPUs
    - **Memory:** 4 GB
    - **Hard Disk:** 42 GB
3. Verify that **Network Adapter 1** is connected to the **VM Network**.
4. Select the **Datastore ISO file** for the **CD/DVD Drive** and enable **Connect**.
5. Review all hardware settings.
6. Click **Next** to continue.

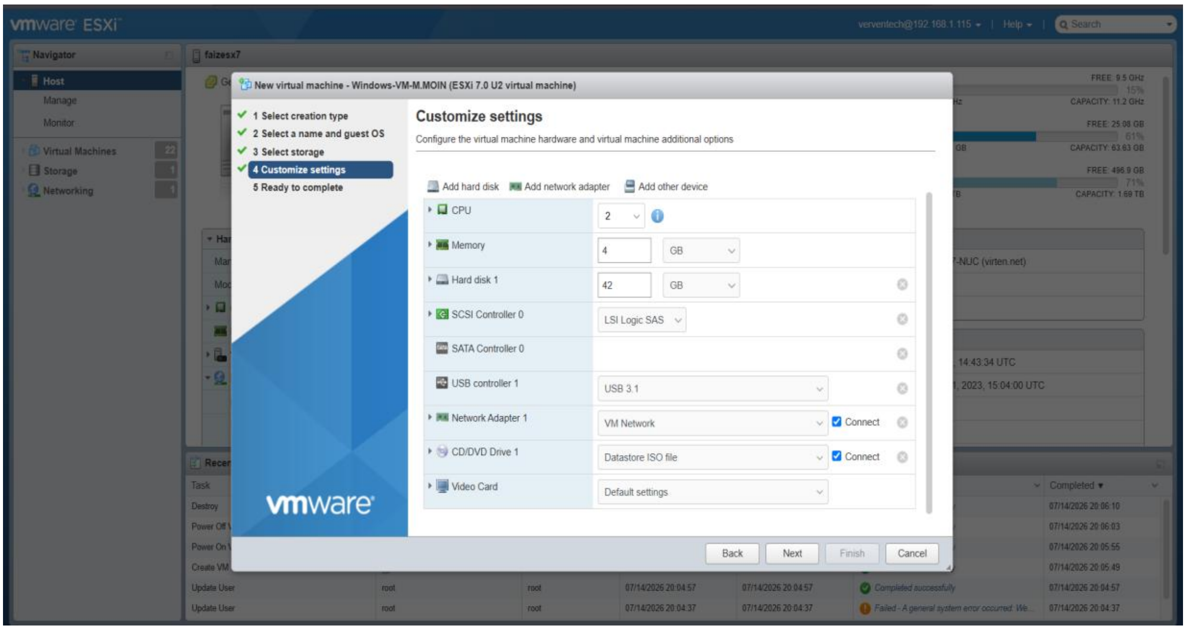


Figure 4: Configuring the virtual hardware settings before virtual machine deployment.

Note: The **Customize Settings** page allows administrators to allocate CPU, memory, storage, networking, and installation media to the virtual machine. Correct hardware configuration ensures a smooth and successful operating system installation.

Step 5: Ready to Complete

Objective

Review the virtual machine configuration before creating the virtual machine.

Procedure

1. Navigate to the **Ready to Complete** page.
2. Review the virtual machine name and guest operating system.
3. Verify the selected datastore.
4. Confirm the configured CPU, memory, storage, and network settings.
5. Ensure the Windows ISO image is attached.
6. Click **Finish** to create the virtual machine.

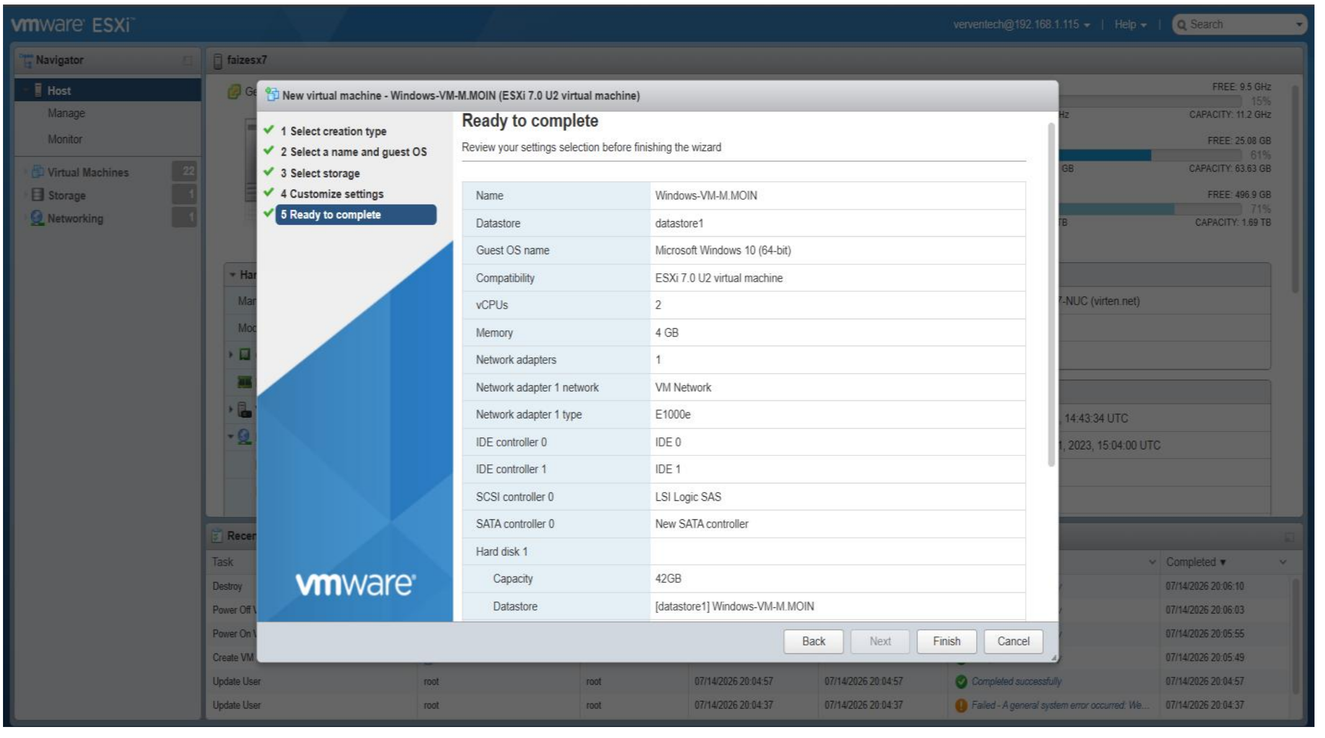


Figure 5: Reviewing the virtual machine configuration before deployment.

Note: The **Ready to Complete** page provides a final summary of all virtual machine settings. Reviewing the configuration before clicking **Finish** helps prevent deployment errors and ensures the virtual machine is created with the intended specifications.

Step 6: Select Installation Type

Objective

Choose the appropriate Windows installation method for a new virtual machine.

Procedure

1. Wait for the Windows Setup wizard to load.
2. On the **Which type of installation do you want?** screen, select **Custom: Install Windows only (advanced)**.
3. This option performs a clean installation of Windows on the new virtual machine.
4. Click **Next** to continue.

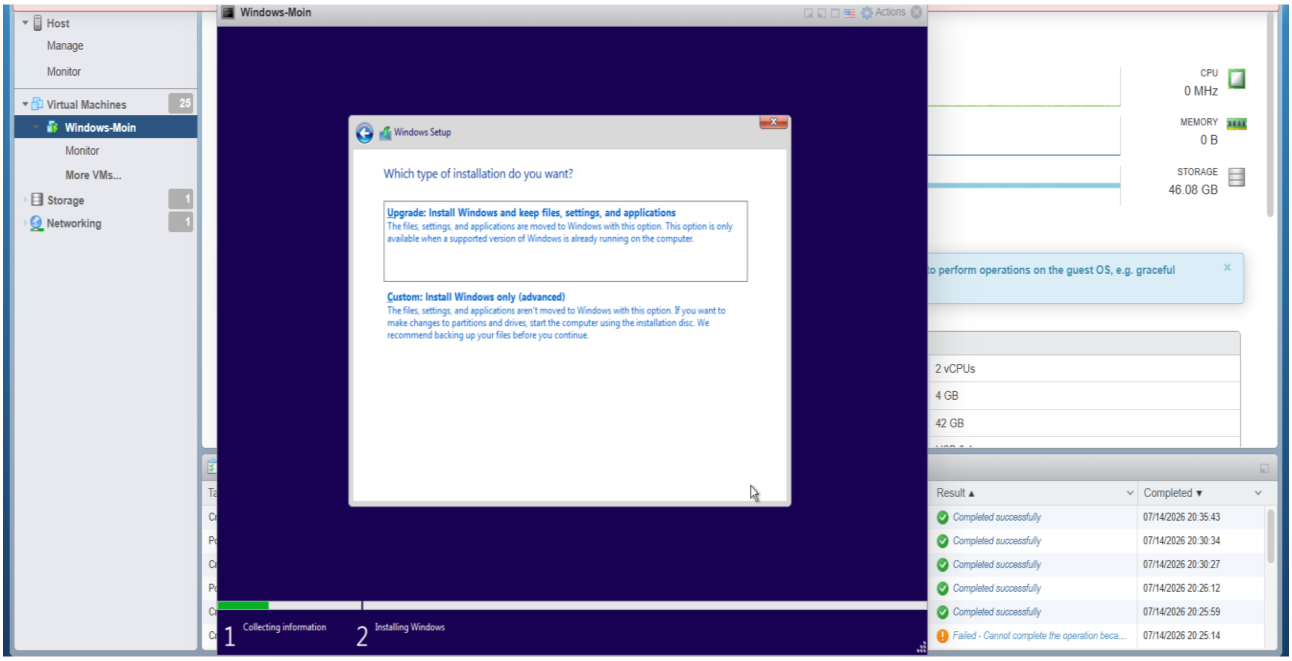


Figure 6: Selecting the **Custom: Install Windows only (advanced)** installation option.

Note: The **Custom: Install Windows only (advanced)** option is recommended for new virtual machines because it installs Windows on a fresh virtual disk without preserving any previous files or settings.

Step 7: Select Installation Drive

Objective

Select the virtual hard disk where the Windows operating system will be installed.

Procedure

1. Wait for the **Where do you want to install Windows?** screen to appear.
2. Select **Drive 0 Unallocated Space**.
3. Verify that the available disk size is **42.0 GB**.
4. Click **Next** to begin the Windows installation.

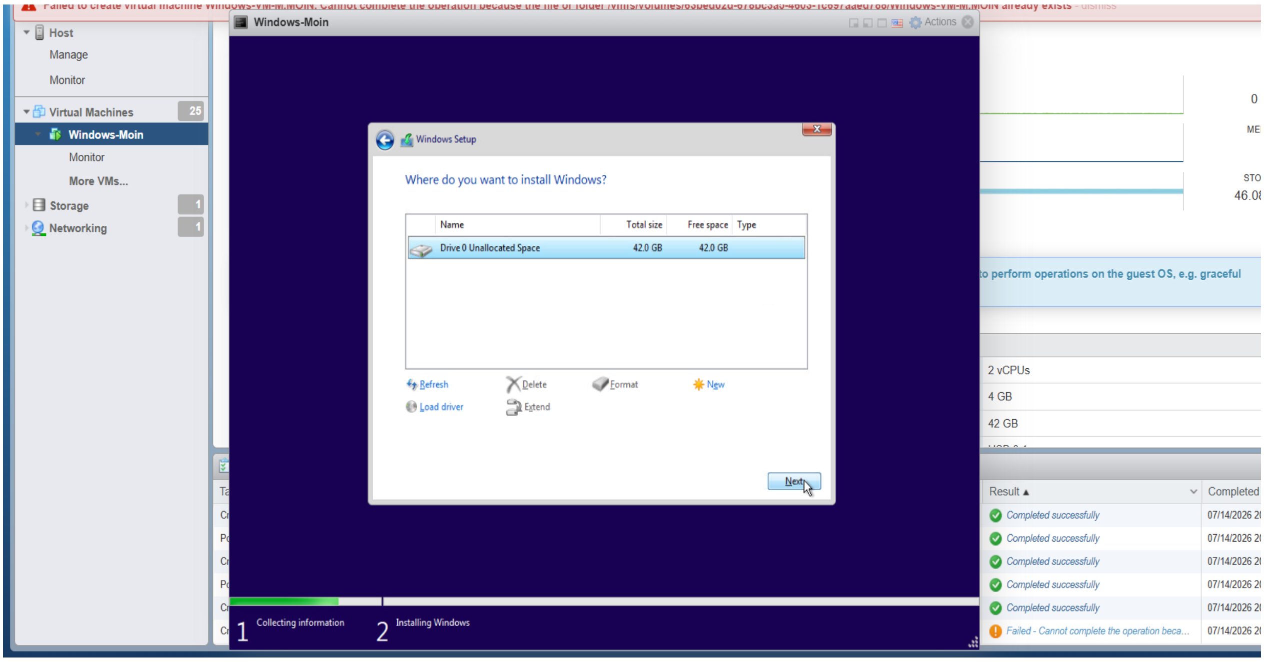


Figure 7: Selecting the unallocated virtual disk for Windows installation.

Note

Note: The **Drive 0 Unallocated Space** represents the newly created virtual hard disk assigned to the virtual machine. Windows automatically creates the required system partitions during installation when the unallocated space is selected.

Step 8: Installing Windows

Objective

Install the Windows operating system on the selected virtual hard disk.

Procedure

1. After selecting the installation drive, click **Next**.
2. Wait while Windows Setup copies the installation files.
3. Allow Windows Setup to prepare the files for installation.
4. Wait as Windows installs features and updates automatically.
5. Do not turn off the virtual machine during this process.
6. The virtual machine will restart automatically after the installation is complete.

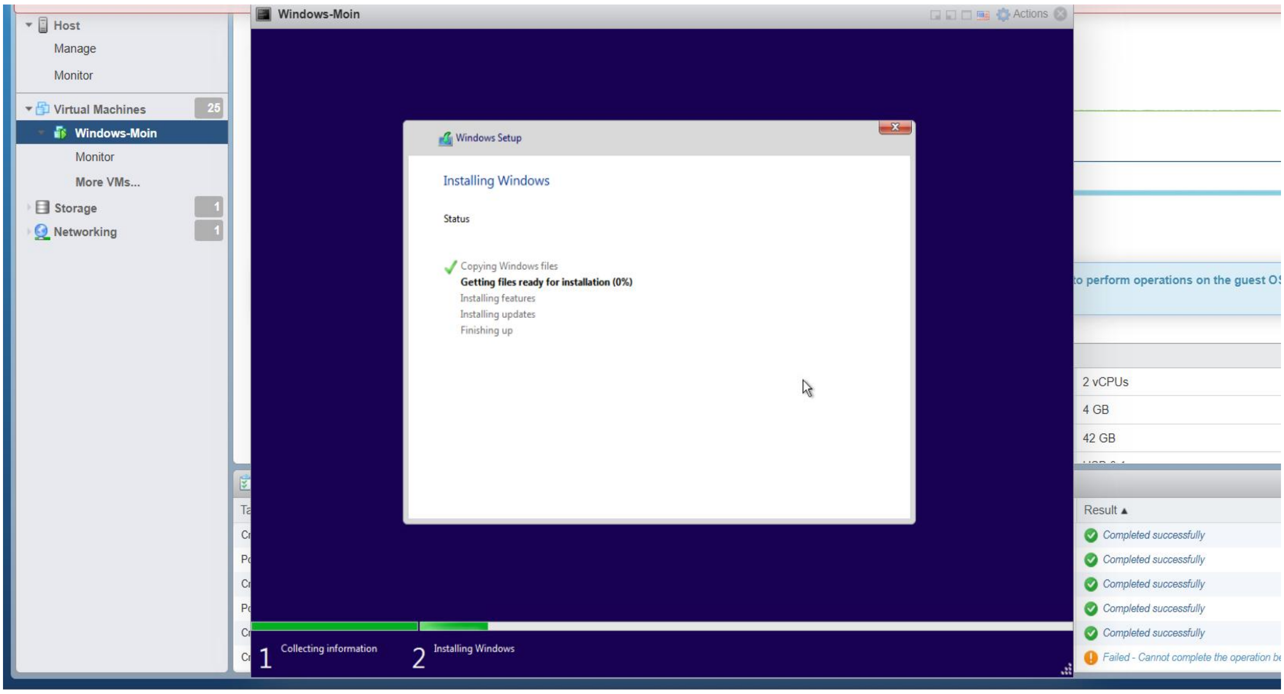


Figure 8: Windows installation in progress.

Note:

The Windows installation process includes copying files, preparing the installation, installing Windows features, installing updates, and completing the setup. Depending on the system resources, this process may take several minutes and the virtual machine may restart automatically.

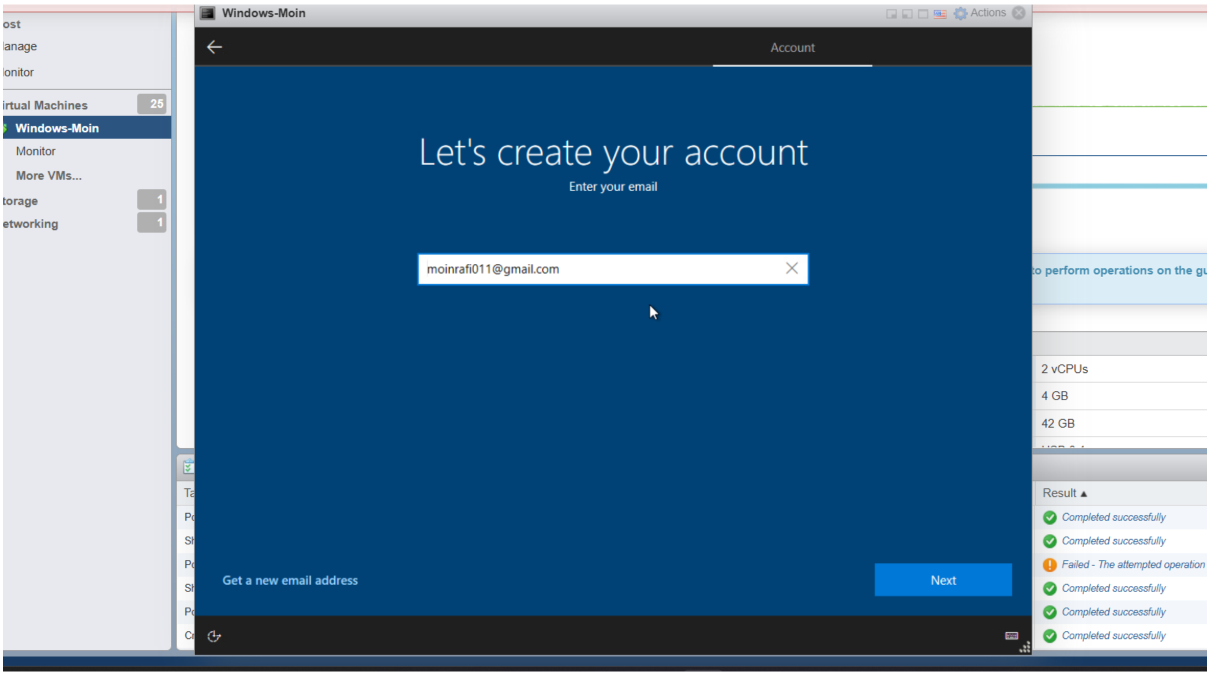
Step 9: Initial Windows Setup

Objective

Complete the initial Windows setup by configuring the user account and basic operating system preferences.

Procedure

1. Wait for Windows Setup to complete the installation and restart automatically.
2. On the **Let's create your account** screen, enter your Microsoft account email address and click **Next**.



3. Enter the account password and sign in.
4. Provide the required personal information, such as your **Date of Birth** and **Region**, if prompted.
5. Create and confirm a **Windows PIN** for secure sign-in.
6. Review the privacy and personalization settings, then continue with the default or preferred options.
7. Wait while Windows completes the final configuration and prepares the desktop.

Note

During the initial Windows setup (OOBE), Microsoft guides the user through account creation, identity verification, security configuration, and privacy preferences. These settings are required only once and prepare the operating system for first-time use.

Step 10: Verifying Successful Installation

Objective

Verify that Windows has been installed successfully and that the virtual machine is ready for use.

Procedure

1. Wait for Windows Setup to complete all installation and configuration tasks.
2. Allow the virtual machine to restart automatically if required.
3. Sign in to Windows using the configured account.
4. Confirm that the Windows desktop loads successfully.
5. Verify that essential desktop components, such as the **Recycle Bin**, **Microsoft Edge**, and the **Start Menu**, are visible.
6. Ensure the virtual machine is responsive and ready for further configuration or software installation.

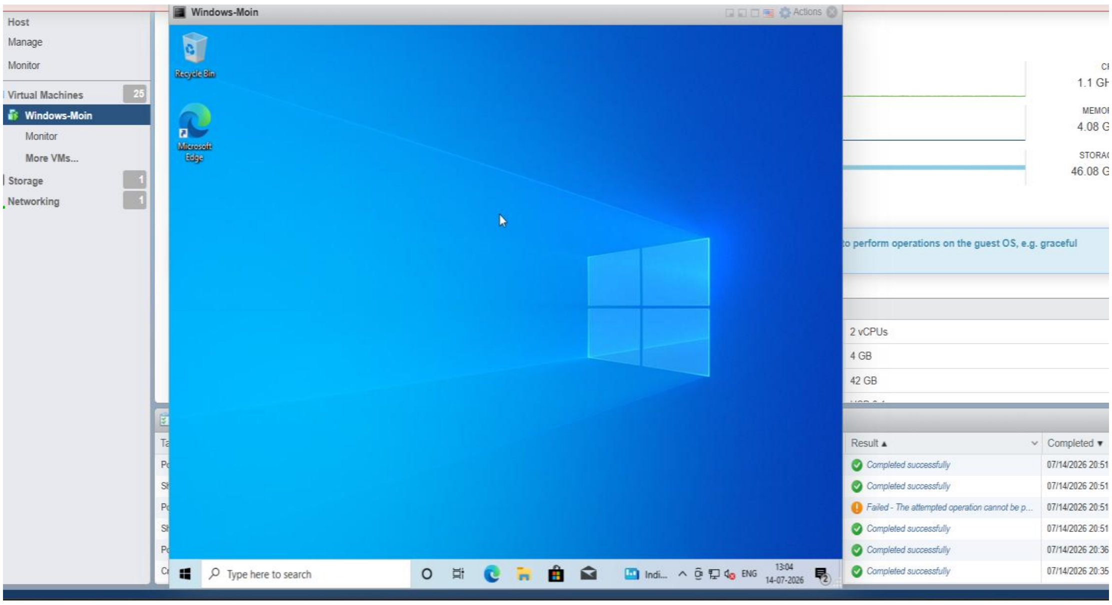


Figure 10: Windows desktop displayed after successful installation.

Note

The appearance of the Windows desktop confirms that the operating system has been successfully installed and configured. At this stage, the virtual machine is ready for additional tasks such as installing VMware Tools, configuring network settings, installing applications, and performing further system customization.